

1. In radio resource management, AI/ML can predict user traffic demand and can assign bandwidth and power to other users who need the most radio resources. AI/ML can analyze network performance data to see if any weird behavior is being detected and help predict equipment failures before it even happens.
2. Self optimization adjusts the network parameters like antenna tilting settings to improve performance. Self healing in SON detects network faults and automatically takes action into correcting the mistakes.
Predictive maintenance uses AI/ML to identify equipment that may fail in the future. This makes results better in better user experience, having lower latency, reduces stress from calls.
3. By normalizing demand, the system can make fair and efficient decisions and also improve the network overall while maintaining the quality of services for all users. This relates to the real-world since Open RAN resource scheduling, bandwidths in network resources are usually limited. The resource scheduling balances many users with different traffic and service priorities over others.
4. One difference is, Traditional RAN's hardware/software are provided by single vendors while Open RAN are usually separate and can be found by different vendors. Another difference is Traditional uses proprietary interfaces while Open uses standard open interfaces.
5. Near-RT RIC operates on a timescale of 10ms to 1 second while Non-RT RIC operates on timescales more than 1 second.
Near-RT handles fast decisions in operating live traffic and Non-RT focuses on more strategy and AI models for the network